

# Drugs for Disorders of the Respiratory System

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# Asthma

- Asthma is a chronic inflammatory disease characterized by
  - Inflammation of the airways
  - Bronchial hyper-reactivity
  - Bronchospasm
- Major symptoms
  - Shortness of breath
  - Cough
  - Chest tightness
  - Wheezing
  - Rapid respiration

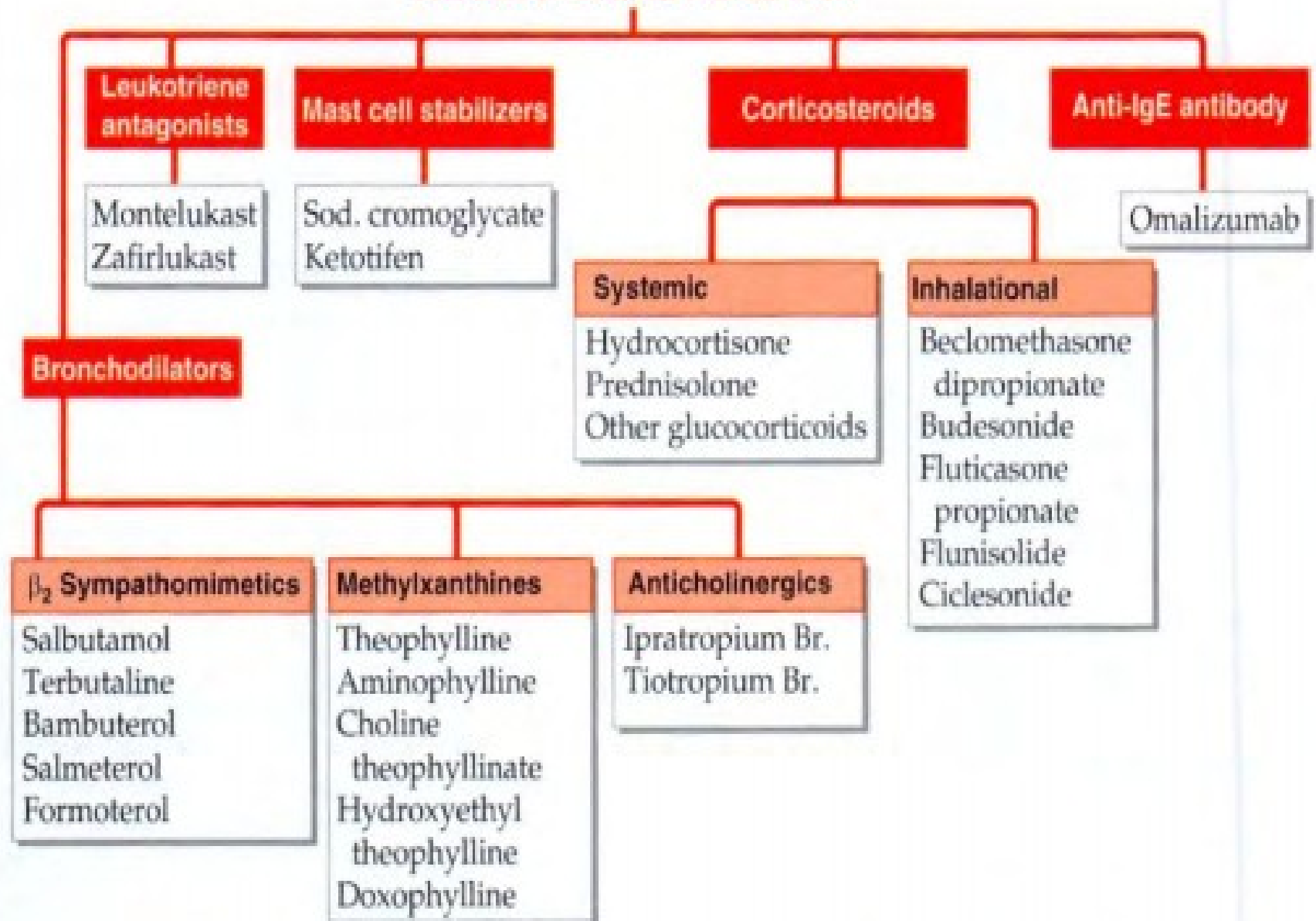
**TABLE 76.5 ■ Classification of Asthma Severity and Recommended Step for Initial Treatment**

|                                                        |                                                                                                                                   | Persistent Asthma                                                                |                                                                                              |                                                                                      |
|--------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|                                                        | Intermittent Asthma                                                                                                               | Mild                                                                             | Moderate                                                                                     | Severe                                                                               |
| CURRENT IMPAIRMENT                                     |                                                                                                                                   |                                                                                  |                                                                                              |                                                                                      |
| Asthma symptoms                                        | ≤2 days/week                                                                                                                      | >2 days/week, but not daily                                                      | Daily                                                                                        | Throughout the day                                                                   |
| Nighttime awakenings                                   | ≤2 times/month                                                                                                                    | 3–4 times/month                                                                  | More than once a week, but not nightly                                                       | Often 7 times/week                                                                   |
| SABA used to control symptoms (but not to prevent EIB) | ≤2 days/week                                                                                                                      | >2 days/week, but not daily, and not more than once on any day                   | Daily                                                                                        | Several times a day                                                                  |
| Effect on normal activity                              | None                                                                                                                              | Minor limitation                                                                 | Some limitation                                                                              | Severe limitation                                                                    |
| Lung function tests                                    | Normal FEV <sub>1</sub> between exacerbations<br>FEV <sub>1</sub> > 80% of predicted<br>FEV <sub>1</sub> /FVC normal <sup>a</sup> | FEV <sub>1</sub> > 80% of predicted<br>FEV <sub>1</sub> /FVC normal <sup>a</sup> | FEV <sub>1</sub> >60% but <80% of predicted<br>FEV <sub>1</sub> /FVC reduced 5% <sup>a</sup> | FEV <sub>1</sub> <60% of predicted<br>FEV <sub>1</sub> /FVC reduced >5% <sup>a</sup> |
| FUTURE RISK                                            |                                                                                                                                   |                                                                                  |                                                                                              |                                                                                      |
| Exacerbations requiring oral glucocorticoids           | 0–1/yr                                                                                                                            | ≥2/yr                                                                            | ≥2/yr                                                                                        | ≥2/yr                                                                                |
| Recommended step for initial treatment <sup>b</sup>    | Step 1                                                                                                                            | Step 2                                                                           | Step 3                                                                                       | Step 4 or 5                                                                          |

# Treatment of asthma

- **Goals of therapy**
  - To decrease the intensity and frequency of asthma symptoms
  - To prevent future exacerbations
  - To minimize limitations in activity related to asthma symptoms
- **Drugs**
  - Anti-inflammatory drugs
  - Bronchodilators
  - Most of them are administered by inhalation

## DRUGS FOR BRONCHIAL ASTHMA



# Bronchodilators

- Provide symptomatic relief in patients with asthma and COPD
- **MoA:** cause bronchodilatation through  $\beta 2$ -receptor stimulation
  - Increased cAMP formation in bronchial muscle cells
    - Relaxation
- Have no anti-inflammatory effects

# $\beta$ 2-agonists

- **Quick relief**
  - Short-acting  $\beta$ 2 agonists (SABAs)
  - Have a rapid onset of action (5 to 30 minutes)
  - Taken as PRN
  - Provide relief for 4 to 6 hours
  - Appropriate for patients with mild, intermittent asthma or exercise-induced bronchospasm
  - Not be used as monotherapy for patients with persistent asthma
  - Includes **albuterol (salbutamol), levalbuterol & terbutaline**

# $\beta$ 2-agonists

- **Long-term control**
  - Long-acting  $\beta$ 2 agonists (LABAs)
  - Providing bronchodilation for at least 12 hours
  - Taken as a fixed schedule
  - Should be used only in combination with an asthma controller medication, such as an inhaled corticosteroid (ICS).
  - Used as adjunctive therapy for attaining control in moderate to severe asthma
  - Include **Salmeterol** and **formoterol**
- Ultralongacting  $\beta$ 2 agonists, such as **indacaterol**, **olodaterol**, **vilanterol**, and **bambuterol**.
  - Mostly for COPD



# $\beta$ 2-agonists

- Adverse effects:
  - Tachycardia/ angina
  - Hyperglycemia
  - Hypokalemia
  - Hypomagnesemia
    - These AEs are minimized with inhaled

# Anti-cholinergic drugs

- Mostly used for COPD
- Ipratropium (short-acting, 4-6 hrs) and tiotropium (long acting, 24 hrs)
- Ipratropium is effective against allergen-induced asthma and EIB,
  - but is less effective than the beta2 agonists
- Tiotropium is approved for maintenance therapy of bronchospasm associated with COPD
- MoA: block muscarinic cholinergic receptors in the bronchi
  - Prevent bronchoconstriction and mucus secretion
- **Adverse effects:** xerostomia and bitter taste

# Methylxanthines: Theophylline

- **MoA** of methylxanthines
  - Release of  $\text{Ca}^{2+}$  from sarcoplasmic reticulum, especially in skeletal and cardiac muscle
  - Inhibition of phosphodiesterase (PDE) which degrades cyclic nucleotides
    - concentration of cyclic nucleotides is increased
      - Bronchodilatation in bronchial muscle
  - Blockade of adenosine receptors

# Pharmacokinetics

- Well absorbed orally
- Distributed in all tissues and 50% plasma protein bound
- Extensively metabolized in liver by demethylation and oxidation primarily by CYP1A2
- Only 10% is excreted unchanged in urine
- The  $t_{1/2}$  in adults is 7- 12 hours
- Children eliminate it much faster ( $t_{1/2}$  3-5 hours) and elderly more slowly

# Adverse effects

- Theophylline has a narrow margin of safety
- Headache, nervousness and nausea
- Cardiac arrhythmias and seizures
- Drug-drug interaction
  - Drugs enhance theophylline metabolism: phenytoin, rifampicin, phenobarbitone
  - Drugs inhibit theophylline metabolism: erythromycin, ciprofloxacin, cimetidine

# Corticosteroids/Glucocorticoids

- Glucocorticoids are the drugs of choice for long-term control of airway inflammation
- Administered usually by **inhalation**, but may also be IV or oral
- Severe persistent asthma may require the addition of a short course of oral or intravenous corticosteroids
- Use in Asthma
  - prophylaxis of chronic asthma
  - persistent asthma

# Glucocorticoids

- **MoA:** Inhibit the release of arachidonic acid through inhibition of **phospholipase A2**
  - thereby producing direct anti-inflammatory properties in the airways
  - **Decreasing** the inflammatory cascade (eosinophils, macrophages, and leukocytes)
  - Decreased edema of the airway mucosa
  - Decrease airway mucus production
  - may increase the number of bronchial beta2 receptors, as well as their responsiveness to beta2 agonists

# Glucocorticoids

- Inhaled glucocorticoids (IGC): Beclomethasone, budesonide, fluticasone, mometasone, triamcinolone
- Oral CS: Methylprednisolone, prednisone, and prednisolone
- Severe exacerbation of asthma (status asthmaticus) may require intravenous **methylprednisolone** or oral **prednisone** to reduce airway inflammation.



# Glucocorticoids

- Adverse effects
  - Oropharyngeal candidiasis
    - Prevention: rinsing mouth after drug use
  - Dysphonia
  - Sore throat
  - Adrenal suppression (high dose, long term use)
  - Slow growth in children and adolescents
  - Bone loss (long term use)
  - Hyperglycemia and peptic ulcer disease (especially in oral )
  - ❖ Inhaled steroids are safe during pregnancy

# Mast cell stabilizers

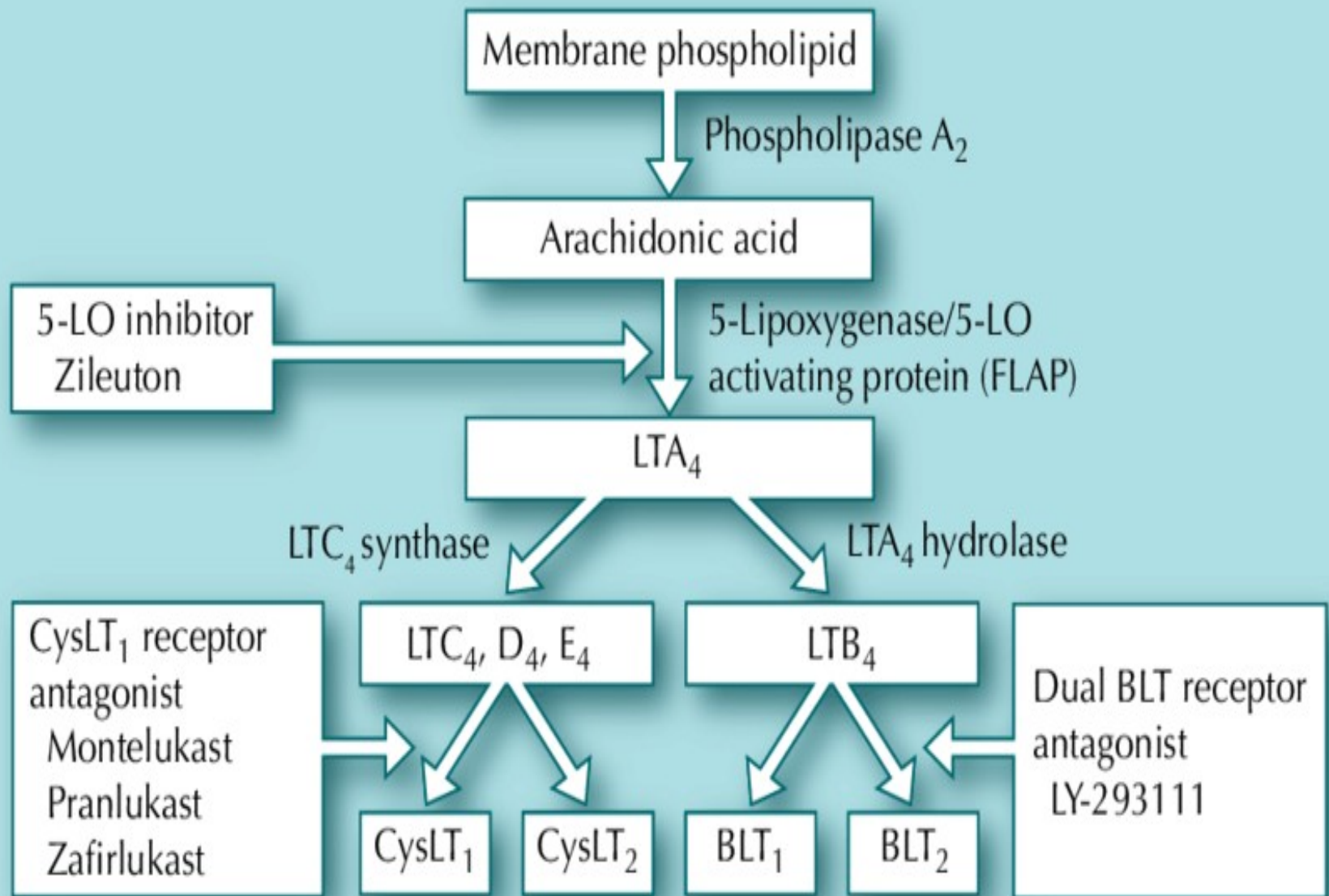
- **Sodium cromoglycate (Cromolyn sod.)**
  - Prophylactic anti-inflammatory agent
  - **Inhibits mast cell degranulation** and release of **histamine, LTs, ILs...**
  - Is **not** a bronchodilator and does not antagonize constrictor action of histamine, Ach, or LTs
  - Therapeutic uses
    - Alternative therapy
    - Exercise-induced bronchospasm
    - Allergic rhinitis

# Mast cell stabilizers

- Pharmacokinetics
  - Not absorbed orally
  - Available as a nebulized solution
  - Has short duration of action
  - Rapidly excreted unchanged in urine and bile
- Adverse effects
  - Cough, throat irritation, and unpleasant taste

# Leukotriene modifiers

- **Zileuton**
  - Inhibitor of **5-lipoxygenase (5-LOX)**
    - Preventing the formation of both LTB<sub>4</sub> and the cysteinyl leukotrienes
- **Zafirlukast and montelukast**
  - Selective antagonist of the cysteinyl leukotriene-1(**cysLT1**) receptor
  - Used for prophylactic therapy of mild-to-moderate asthma as alternatives to inhaled glucocorticoids.
  - In severe asthma, they have additive effect with inhaled steroids, may permit reduction in steroid dose and need for rescue **β2**-agonist inhalations.
  - Effective in **aspirin-induced** asthma and **exercise** induced asthma.



# Leukotriene modifiers

- **Pharmacokinetics**
  - Orally active and highly protein bound.
  - Undergo extensive hepatic metabolism
  - Food reduces zafirlukast absorption
  - The plasma  $t_{1/2}$  of montelukast is 3-6 hours, while that of zafirlukast is 8-12 hours, & zileuton is 2.5 hours.
  - **Zileuton** and its metabolites are excreted in urine
  - **Zafirlukast, montelukast**, and their metabolites undergo biliary excretion

# Leukotriene modifiers

- **Adverse effects**
  - **Neuropsychiatric** effects (depression, anxiety, abnormal dreams, hallucinations, insomnia, and suicidal thinking)
  - **Zafirlukast**: headache and GI disturbance
  - Zileuton: hepatotoxic and inhibitor of CYP1A2
  - Zafirlukast is an inhibitor of CYP2C8, 2C9, and 3A4

# Monoclonal antibodies

- Omalizumab
  - Selectively binds to human immunoglobulin E (IgE)
    - Decreased binding of IgE to its receptor on the surface of mast cells and basophils
  - No benefit has been noted in non-allergic asthma
  - Reserved for resistant asthma patients with positive skin tests or raised IgE levels
- Mepolizumab, benralizumab, and reslizumab
  - Are interleukin-5 (IL-5) antagonists



# Monoclonal antibodies

- Indicated for the treatment of severe persistent asthma in patients who are poorly controlled with conventional therapy
- Their use is limited by the:
  - High cost
  - Route of administration (IV for reslizumab and SC for others)
  - Adverse effect profile
- Adverse effects:
  - Anaphylactic reaction (rare)
  - Arthralgias, fever, rash
  - Increased risk of infections

**TABLE 76.11 ■ Stepwise Approach to Managing Asthma**

|                                      | Long-Term Control Drugs (Taken Daily)                                                     |                                                                      | Quick-Relief Drugs<br>(Taken PRN) |
|--------------------------------------|-------------------------------------------------------------------------------------------|----------------------------------------------------------------------|-----------------------------------|
|                                      | Preferred                                                                                 | Alternative                                                          |                                   |
| ADULTS AND CHILDREN AGE 12 AND OLDER |                                                                                           |                                                                      |                                   |
| Step 1                               | No daily medication needed                                                                |                                                                      | SABA                              |
| Step 2                               | Low-dose IGC                                                                              | Cromolyn, LTRA, or theophylline                                      | SABA                              |
| Step 3                               | Low-dose IGC + LABA<br>OR<br>Medium-dose IGC                                              | Low-dose IGC + either LTRA, theophylline, or<br>zileuton             | SABA                              |
| Step 4                               | Medium-dose IGC + LABA                                                                    | Medium-dose IGC + either LTRA, theophylline,<br>or zileuton          | SABA                              |
| Step 5                               | High-dose IGC + LABA                                                                      |                                                                      | SABA                              |
| Step 6                               | High-dose IGC + LABA + oral glucocorticoid                                                |                                                                      | SABA                              |
| CHILDREN AGE 5–11 YEARS              |                                                                                           |                                                                      |                                   |
| Step 1                               | No daily medication needed                                                                |                                                                      | SABA                              |
| Step 2                               | Low-dose IGC                                                                              | Cromolyn, LTRA, nedocromil, or theophylline                          | SABA                              |
| Step 3                               | Low-dose IGC + either LABA, LTRA, or theophylline<br>OR<br>Medium-dose IGC                |                                                                      | SABA                              |
| Step 4                               | Medium-dose IGC + LABA                                                                    | Medium-dose IGC + either LTRA or theophylline                        | SABA                              |
| Step 5                               | High-dose IGC + LABA                                                                      | High-dose IGC + either LTRA or theophylline                          | SABA                              |
| Step 6                               | High-dose IGC + LABA + oral glucocorticoid                                                | High-dose IGC + either LTRA or theophylline +<br>oral glucocorticoid | SABA                              |
| CHILDREN AGE 0–4 YEARS               |                                                                                           |                                                                      |                                   |
| Step 1                               | No daily medication needed                                                                |                                                                      | SABA                              |
| Step 2                               | Low-dose IGC                                                                              | Cromolyn or montelukast                                              | SABA                              |
| Step 3                               | Medium-dose IGC                                                                           |                                                                      | SABA                              |
| Step 4                               | Medium-dose IGC + either LABA or montelukast                                              |                                                                      | SABA                              |
| Step 5                               | High-dose IGC + either LABA or montelukast                                                |                                                                      | SABA                              |
| Step 6                               | High-dose IGC + either LABA or montelukast<br>(Low-dose oral glucocorticoids, if needed.) |                                                                      | SABA                              |

IGC, Inhaled glucocorticoid; LABA, long-acting beta<sub>2</sub> agonist; LTRA, leukotriene receptor antagonist; SABA, short-acting beta<sub>2</sub> agonist.

# Pharmacotherapy of cough and nasal decongestant

- Cough is a protective reflex, its purpose being expulsion of respiratory secretions and foreign particles from air passages.
- It may be productive or non-productive
- Before treating cough, identification of its cause is important.

# Pharmacotherapy of cough

- **Expectorants**
  - Increase bronchial secretion or reduce its viscosity, facilitating its removal by coughing.
  - Example: **Guaifenesin**
- **Mucolytics**
  - Reacts directly with mucus to make it less viscous
  - Eg. hypertonic saline and acetylcysteine
- **Antitussives**
  - Suppress cough centrally and peripherally
  - Should be used **only** for dry **non-productive** cough
  - Classified as opioid and non-opioid antitussives

# Opioid antitussives

- **Codeine**
  - **Decreases** the sensitivity of cough center in the CNS to peripheral stimuli and decrease the frequency and intensity of cough
  - Its therapeutic effects occur at doses lower than those required for analgesia.
  - At higher doses respiratory depression and drowsiness can occur, especially in **children**
  - Adverse effects: constipation, dysphoria, fatigue, & **addiction**.

# Opioid antitussives

- **Pholcodine**
  - It has practically no analgesic or addicting property
  - Similar in efficacy as antitussive to codeine
  - Longer acting-acts for 12 hours

# Nonopioid antitussives

- **Dextromethorphan**
  - A synthetic central N-methyl D-aspartate (NMDA) receptor antagonist
  - Does not depress mucociliary function of the airway mucosa and is practically devoid of constipating action.
  - **Adverse effects:** dizziness, nausea, drowsiness; at high doses hallucinations and ataxia may occur.

# Adjuvant antitussives

- **Antihistamines**
  - Added to antitussive/expectorant formulations
  - First-generation antihistamines are commonly used
    - Chlorpheniramine
    - Diphenhydramine
    - Promethazine
- **Bronchodilators:** Salbutamol and Terbutaline



# Drugs used to treat allergic rhinitis

- Allergic rhinitis is an inflammatory disorder that affects the upper airway.
- Major symptoms are sneezing, rhinorrhea (runny nose), pruritus (itching), and nasal congestion
  - caused by dilation and increased permeability of nasal blood vessels.
- Allergic rhinitis has two major forms: seasonal and perennial
- Treatment includes:
  - Antihistamines
  - Corticosteroids
  - $\alpha$ -Adrenergic agonists

# Antihistamines

- H1 receptor antagonists
  - The second-generation antihistamines (**fexofenadine**, **loratadine**, desloratadine, cetirizine) are mostly used.
  - Topical intranasal antihistamines include olopatadine and azelastine.
    - Intranasal antihistamines provide increased delivery of the drug with fewer adverse effects.

# Corticosteroids

- Intranasal corticosteroids, such as beclomethasone, budesonide, fluticasone, ciclesonide, mometasone, and triamcinolone.
- Onset of action that ranges from 3 to 36 hours.
- For patients with chronic rhinitis, improvement may not be seen until 1 to 2 weeks after starting therapy.
- Adverse effects: nasal irritation, nose bleed, sore throat, and candidiasis (rarely).

# $\alpha$ -Adrenergic agonists

- Nasal decongestants
- Activate  $\alpha_1$ -adrenoceptor in the arterioles of nasal mucosa
  - Vasoconstriction, which leads to decreased edema and reduce airway resistance.
- Includes: phenylephrine, pseudoephedrine and oxymetazoline
- Intranasal formulations should be used for no longer than 3 days due to the risk of **rebound** nasal congestion
- Oral formulations results in a longer duration of action but also increased systemic effects, such as increased blood pressure and heart rate, excitation, & **abuse (pseudoephedrine)**

**Thank you!**